1. (a) Group the following terms into pairs, so that in each pair a quantity is given followed by its corresponding unit.

power, volt, watt, magnetic flux, magnetic flux density, potential difference, tesla, weber. [2]

(b) Explain why it is technically incorrect to define speed as distance travelled per second. Include in your answer the correct statement defining speed. [2]

(c) An archer fires an arrow with an initial velocity v and hits a target which is a distance d away and on the same horizontal level as the bow, as illustrated in **Fig. 1**. The arrow is aimed so that, on release, it makes an angle θ with the horizontal.

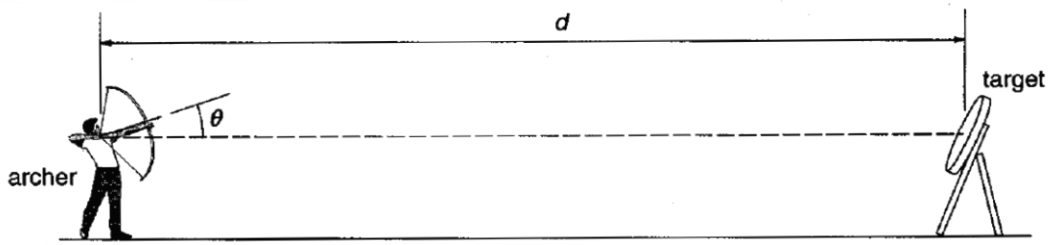

Fig. 1

(i) Assuming air resistance is negligible, write down an expression for d in terms of v, θ, and the time of flight t. [1]

(ii) Write down an expression for the time of flight t in terms of v, θ and the acceleration of free fall g. [1]

(iii) The distance d is given by the expression

$$d = \frac{v^2 \sin 2\theta}{g}.$$

Calculate two possible angles, θ_1 and θ_2, for an arrow with initial velocity $v = 32\ \text{m s}^{-1}$ and a target at a distance d of 94 m from the bow. [3]